2026-06-09

Climatology prediction

https://github.com/tglanz/climatology/tree/main/ml

Context

Given a sequence of K **step diagnostic** ζ, instead of attempting to predict the state of the system at some later time t we attempt to predict a **climatology diagnostic** $[\bar{u}]$.

By doing so, we eliminate the stochastic dimension of the forcing (Stirring in our case).

Window Selection

- Window size K
- Start / End / Stride?

Climatology Range

What is the time frame to compute $[\bar{u}]$ for?

Learnability

The underlying heuristic is that the diagnostic evolution within a window determines the climatology

i.e. for different windows $W_{i,j}$

$$pred(W_i) = pred(W_j) = [\tilde{\bar{u}}]$$

- We don't encode time in the model

Spinup and convergence

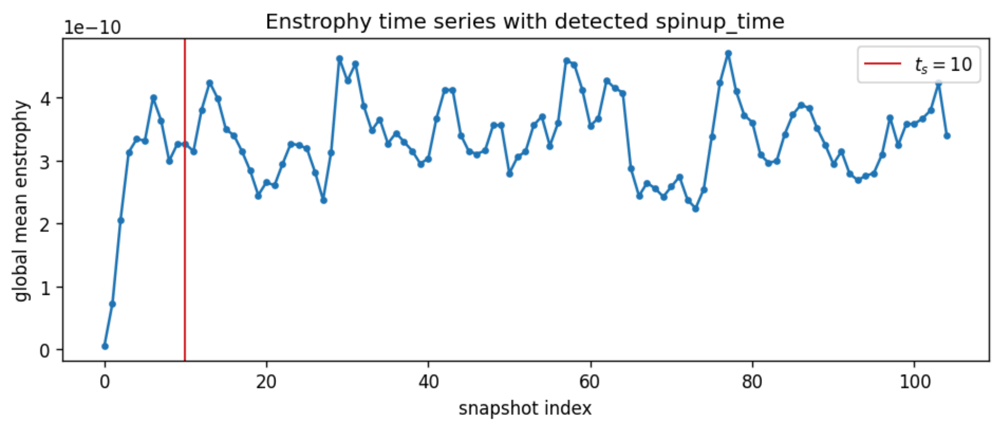
A simulation S is composed of M snapshots; Each snapshot holding dimensions (lons, lats, time, ...) and variables (vor, stirring, ucomp, ...).

- Windows are taken between the **Spinup Time** t_s and the **Convergence Time** t_c
- Climatology $[\bar{u}]_{t_c}^{t_M}$ is computed since t_c to the end of simulation

Spinup

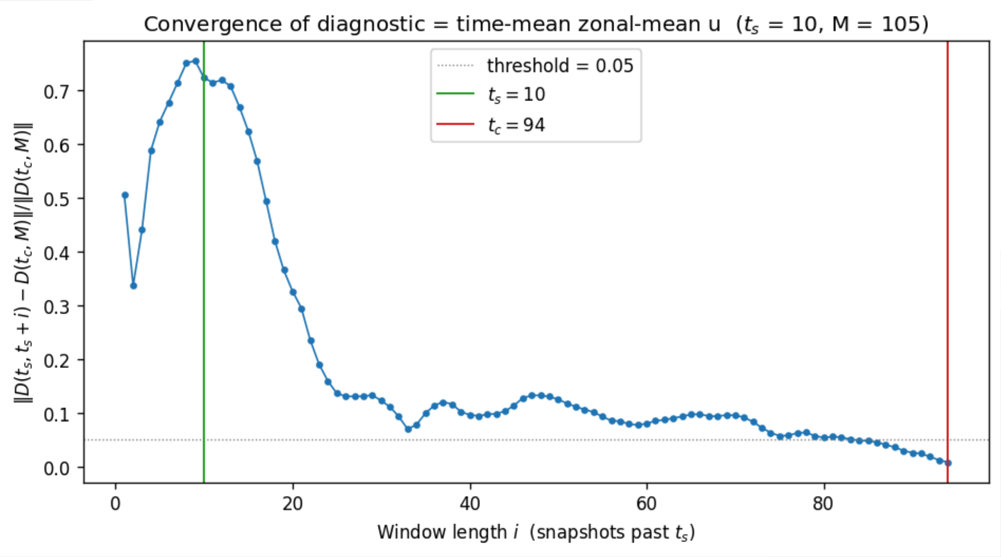
Spinup time is the time t_s after which the simulated dynamics have stabilized post initial conditions

t_c is determined by taking multiple subsequent enstorophy windows and thresholding their change of z-score



Convergence (1/2)

Convergence time is the time $t_c \geq t_s$ after which the time mean has settled similar to the global time mean



Convergence (2/2)

We determine t_c based on the error:

$$\text{err}(i) = \frac{\left| [\bar{u}]_{t_s}^{t_{s+i}} - [\bar{u}]_{t_s}^{t_M} \right|_{\cos}}{\left| [\bar{u}]_{t_s}^{t_M} \right|_{\cos}}$$

Then $t_c = t_s + i^*$, where i^* is the first index such that $\text{err}(i^* - h), \ldots, \text{err}(i^*) < \tau$ for hold h and threshold τ.

Training state